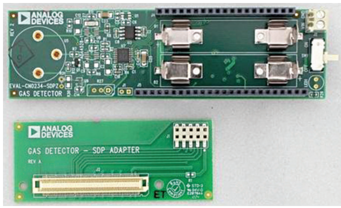
5 Interesting REFERENCE DESIGNS

Title	Portable, Battery-Operated, Carbon Monoxide Sensor	Multi-Connectivity Asset Tracker
Image		
Key Attraction	The reference design features an electrochemical sensor and an adjustable rheostat which enables the user to use the same sensor to detect various gasses. This reference design can work for several months with a couple of AAA cells	This reference design can help you build an ultra-low-power asset tracker that can enable you to track and monitor livestock, vehicles, etc. The reference design features a multi-connectivity asset tracking solution that will improve the accuracy of the location. This reference design also features multiple environmental sensors that can provide complete details about the environment of the asset.
Highlights	CN0234 from Analog Devices is a reference design for a carbon monoxide detector. The design is based on Alphasense CO-AX sensor which detects carbon monoxide in the environment. It is an ultra-low-power design that can run for an extended period on a single supply. The reference design employs an electrochemical sensor and a programmable rheostat that enables the user to detect different gasses by changing the resistance. The reference design can work for several months with a couple of AAA cells. Moreover, it can be upgraded to work with a dual power source, therefore, extending its runtime. This design can detect levels up to 4000 ppm of CO and 0 ppm to 400 ppm within 30 seconds of starting. The reference design uses dual op-amps to achieve lower noise and better accuracy. When connected to an ADC and a microcontroller, or a microcontroller with a built-in ADC, the battery life of this reference design can be from over six months to over one year.	The reference design from ST Microelectronics is a multi-connectivity asset tracking solution with MEMS environmental and motion sensors and Global Navigation Satellite System (GNSS) positioning that targets advanced tracking applications such as livestock monitoring, fleet management, and logistics. It has a modular design and offers two wireless SoCs for enabling both short- and long-range communication. Moreover, the reference design features a host of environmental and motion sensors including temperature sensor, pressure sensor, humidity sensor, accelerometer, and gyroscope. The sensors enable the reference design to monitor and log the environmental conditions to detect any damage caused due to harsh environment and force applied on the asset. The reference design includes an optimised power management which enables long battery life. It also comes with an asset tracking mobile application which is available for both Android and IoS users. The design comes with both an SMA antenna and NFC antenna which improves the accuracy of the design. Housed in a plastic case the reference design has an operational temperature from +5 to 35°C.
Applications	The module is suitable for use as gas detector in kitchens, garages, indoor parking lots	• Fleet management • Livestock monitoring • Goods monitoring • Logistics
OEM Brand	Analog Devices	ST Microelectronics
URL	https://www.electronicsforu.com/electronics-projects/electronics-design-guides/reference-design-of-a-portable-battery-operated-carbon-monoxide-sensor	https://www.electronicsforu.com/electronics-projects/electronics-design-guides/reference-design-for-a-multi-connectivity-asset-tracker
Short URL	https://qrco.de/CO_sensor	https://qrco.de/asset_tracker
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